

Date

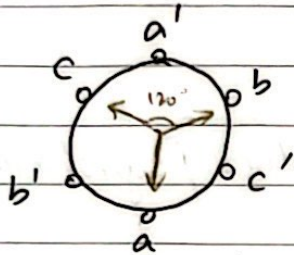
DAY-17

Synch Motor :: (same M/C as synch gen)

stator (armature) 3ϕ

Rotor (Field/Excitation) DC

Let us consider 3ϕ armature (ignoring rotor)
(3ϕ current)



$$B_{aa'} = B_m \sin \omega t \angle 0^\circ T$$

$$B_{bb'} = B_m \sin (\omega t - 120) \angle 120^\circ T$$

$$B_{cc'} = B_m \sin (\omega t - 240) \angle 240^\circ T$$

$$B_{net} = B_{aa'} + B_{bb'} + B_{cc'}$$

Diff. cases of B_{net}

$$\omega t = 0^\circ$$

$$B_{net} = 1.5 B_m \angle -90^\circ (-1.5 B_m \hat{y})$$

$$\omega t = 90^\circ$$

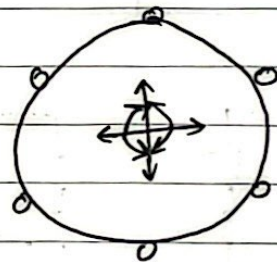
$$B_{net} = 1.5 B_m \angle 0^\circ (1.5 B_m \hat{x})$$

$$\omega t = 180^\circ$$

$$B_{net} = 1.5 B_m \angle 90^\circ$$

$$\omega t = 270^\circ$$

$$B_{net} = 1.5 B_m \angle 180^\circ$$

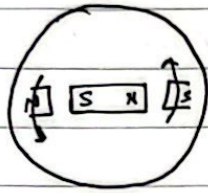


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FI-YAO

If we supply 3 ϕ current to a 3 ϕ winding, a rotating magnetic field is produced.

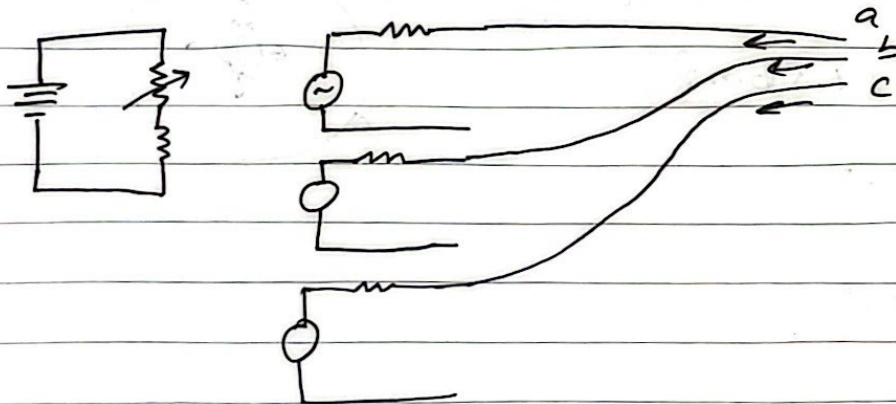
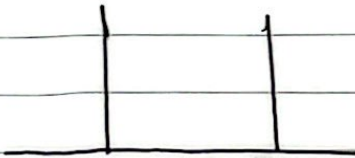
Rotating Mag. Field ..



Synch motor operates when rotor poles lock/synchronize with stator poles.

$$f_e = \frac{n_{syn} P}{120}$$

$$n_{synch} = \frac{f_e 120}{P}$$



$$V_T = E_A + I_A (j X_s)$$

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$$f_c = 50 \quad P = 2$$

$$n = \frac{50(120)}{2}$$

$$= 3000 \text{ rpm}$$

$$3000 \times \frac{2\pi}{60}$$

$$50 \text{ rev./sec}$$

starting synch. Motor

- 1) Use a prime Mover. 2990 rpm
- 2) Run as gen/ make it // with supply, switch off prime over.
- 3) Use of variable freq. prime
- 4) Use damper winding

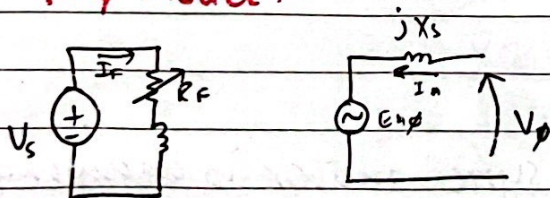
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Synch. Motor :- (synch. Machine)

Rotating mag. field

Locking of rotor field with rotating mag. field

1- ϕ Model :-



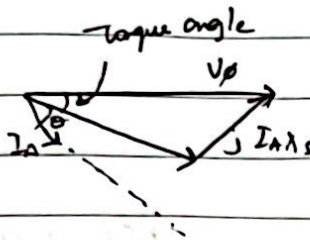
$$V_\phi = E_A + I_A(jX_s)$$

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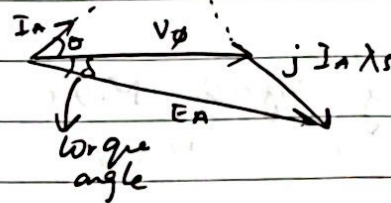
Now self starting motor
Fixed speed motor $f_e = \frac{n_{\text{sync}} P}{120}$

operate in a wide range of power factors
(lagging $\rightarrow$ leading)

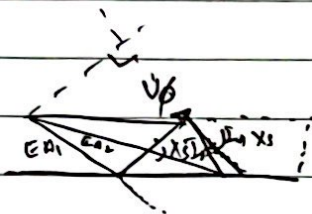
S M with lagging PF



Leading PF



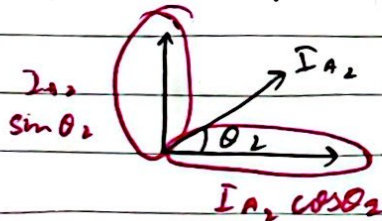
Operation of S M at const load and variable I_f
 I_f & E_A



81-VAC

If I_f increases

θ of I_a goes from lagging towards leading



QUESTION:-

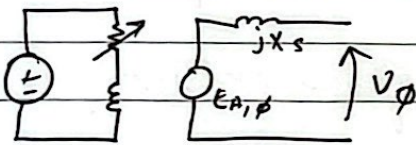
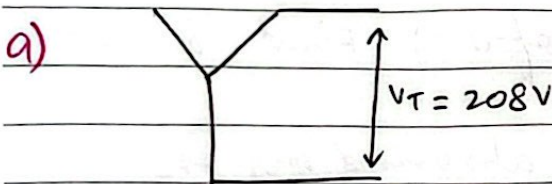
V-connected, 6 pole sync. motor is drawing 40A at 0.8 pf lagging at rated voltage. The field current under

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This condition is 2.7 A . And the synch. reactance is 0.9Ω . Assume a LOCC and find out the following

- torque angle and internally generated voltage
- If same motor is arranged, same output power, what would be internally generated voltage & torque angle at unity power.
- Find out the field current for part (b)
- Assuming same motor at leading pf of 0.7 . Find angle and field current.
- Find out the reactive power delivered or absorbed by the motor for all the above cases.
- Find speed of motor and output torque of the motor.

Solution:-



$$E_{A,\phi} = ?$$

$$V_\phi = \frac{208}{\sqrt{3}} = 120.1 \text{ V}$$

$$I_A = 40 \angle -\cos^{-1} 0.8 = 40 \angle -36.87^\circ \text{ A}$$

$$E_{A,\phi} = 120.1 + (40 \angle -36.87^\circ)(j0.9)$$
$$= 102.66 \angle -16.3^\circ \text{ V}$$

E_A δ (torque angle)

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b) $\cos \theta_2 = 1$

$$I_{A2} \cos \theta_2 = I_{A1} \cos \theta_1$$

$$A_3 \text{ Pord} = \text{const}$$

$$I_{A2}(j) = 40(0.8)$$

$$|I_{A2}| = 32 \text{ A}$$

$$I_{A2} = 32 \angle 0^\circ$$

$$I_{A2} = 123.56 \angle -13.48^\circ$$

$\downarrow$ E_A $\downarrow$ δ

c) $\frac{E_{A2}}{I_{F2}} = \frac{E_{A1}}{I_{F1}}$ (linear DCC)

$$\frac{123.56}{I_{F2}} = \frac{102.66}{2.7}$$

$$I_{F2} = 3.24 \text{ A}$$

d) Assuming same motor at leading pf of 0.7. Find δ and fild current

e) Find out the reactive power delivered or absorbed by the motor for all the above cases

f) Find speed of motor and output torque of the motor

d) $\cos \theta_3 = 0.7$ leading

$$I_{A3} \sin \theta_3 = I_{A1} \cos \theta_1$$

$$I_{A3}(0.7) = 40(0.8)$$

$$I_{A3} = 45.7 \text{ A}$$

$$I_{A3} = 45.7 \angle 45.56^\circ$$

$$E_{A3} = 152.2 \angle -10.9^\circ$$

$$\delta_3 = -10.9^\circ$$

$$\frac{E_{A3}}{I_{F3}} = \frac{E_{A1}}{I_{F1}}$$

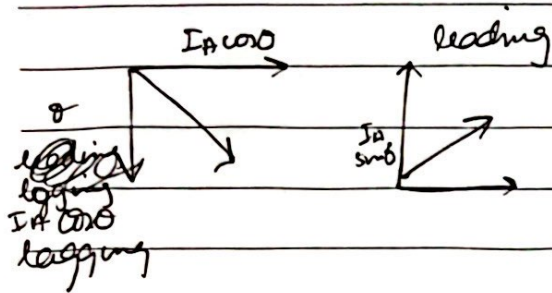
$$I_{F3} = 2.7 \text{ A}$$

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$$I_{F3} = 4 \text{ A}$$

e) $S = \sqrt{3} V I$
 $P = S \cos \theta$

$S = 70 \text{ kVA}$
 $\text{PF} = 0.2, 0.4, 0.6, 0.8 \text{ lag}$

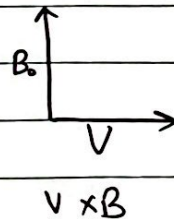
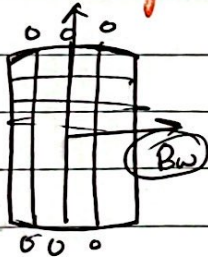


$P_1 = 2 \text{ kW}$
 $P_2 = 4 \text{ kW}$
 $P_3 = 6 \text{ kW}$
 $P_4 = 8 \text{ kW}$

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$$f_c = \frac{n_s \sin \theta}{120} P$$

Starting SM using damper winding :-



If we have a 3 ϕ stator and a rotor with short circuited winding, this motor shall start and run automatically when energized.

this is the concept of induction motor.

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IS-YAT

$$V_{cx} = 5.2$$

$$V_c = V_D + \vec{\omega}_c \times \vec{r}_{c/D}$$

$$\begin{bmatrix} 5.2 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \omega_c \end{bmatrix} \times \begin{bmatrix} 0 \\ -0.1 \\ 0 \end{bmatrix}$$

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Induction Motor :-

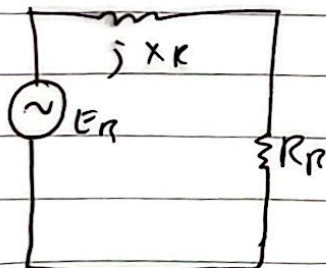
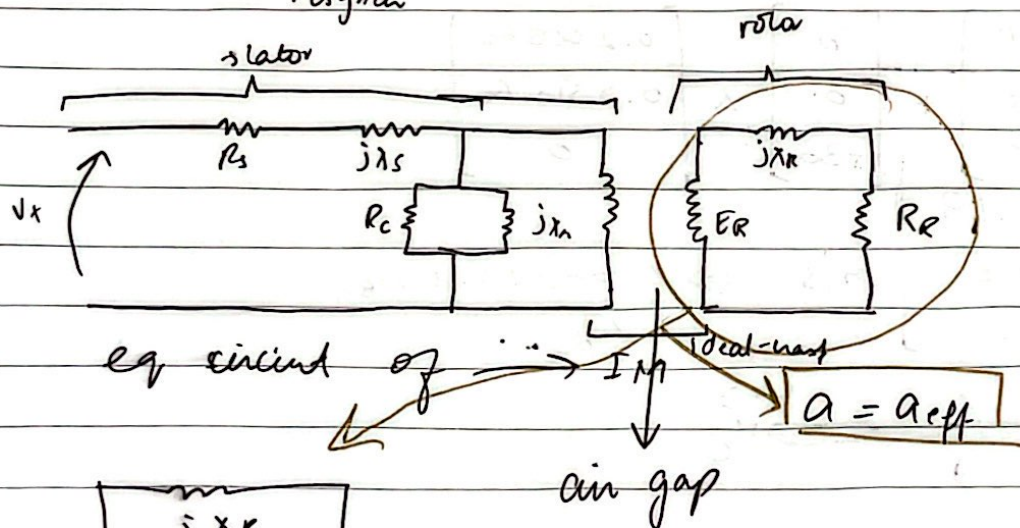
$$n_{sync} = \frac{120 f_e}{P}$$

n_m = motor's speed ($n_m < n_{sync}$)

$$n_{slip} = n_{sync} - n_m$$

slip of motor. (s)

$$s = \frac{n_{sync} - n_m}{n_{sync}}$$



$$I_R = \frac{E_R}{R_R + jX_R}$$

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Sheza
Sheza
Sheza
Sheza
Sheza

$$X = 2\pi f L$$

$$X_R = 2\pi f_r L$$

$f_r = \text{motor frequency}$

$$f_r = s f_e$$

$$X_R = 2\pi (s f_e) L_R$$

$$X_R = s (2\pi f_e L_R)$$

$$X_R = s X_{R0}$$

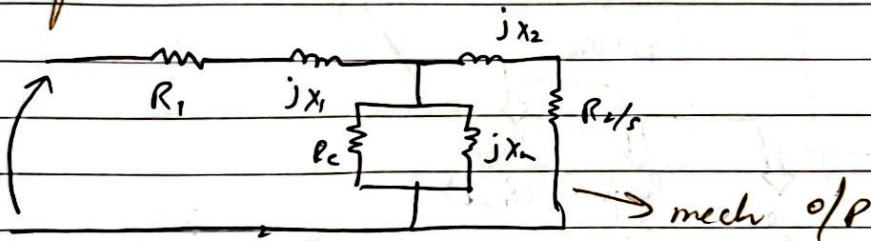
$$E_R = s E_{R0}$$

$$I_R = \frac{s E_{R0}}{R_R + j s X_{R0}}$$

$$= \frac{s E_{R0}}{s \left(\frac{R_R}{s} + j X_{R0} \right)}$$

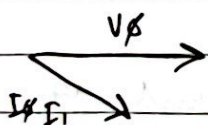
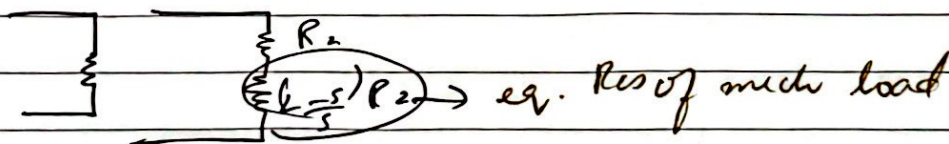
$$= \frac{E_{R0}}{\frac{R_R}{s} + j X_{R0}}$$

Shift to stator side



$$X_2 = a_{eff} X_{R0}$$

$$R_2 = a_{eff} (R_R)$$



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$$P_{in} = \sqrt{3} V_L I_L \cos \theta$$

$$= 3 V_{\phi} I_{\phi} \cos \theta$$

$$P_{scl} = 3 I^2 R$$

$$I_L = I_{\phi}$$

$$P_{AG} = P_{in} - (P_{scl} + P_{wle})$$

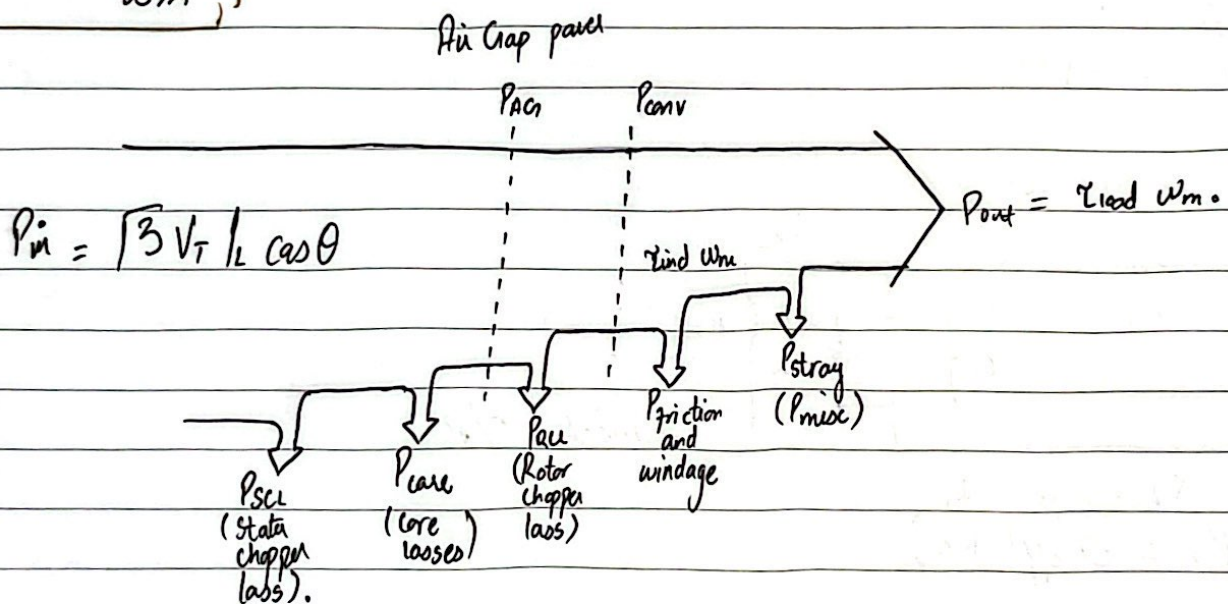
$$P_{PCL} = 3 I_r^2 R_r$$

$$P_{conv} = P_{AG} - P_{PCL}$$

$$P_{out} = P_{conv} - (P_{mech} + P_{misc})$$

$$\tau_{ind} = \frac{P_{conv}}{\omega_m}$$

$$\tau_{out} = \frac{P_{out}}{\omega_m}$$



Question: (On Camera).

Extra Explanation add to question.

Assume that for the above mention motor total losses are

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1200 watts. Find at the input line current assuming PF of 0.7 lagging. Also find at the total power absorbed by the motor.

Question :

A 50hp, 400 volt, 4 pole, 50 Hz Δ connected motor is running at a speed of 1425 rpm. Assuming that the PF is 0.8 lagging and total losses are 3kW find out.

- i) Slip speed
- ii) Slip
- iii) Output torque
- iv) line current
- v) Reactive power absorbed by the motor.

Induction Motors

$$1. \quad n_{sync} = \frac{120 f_e}{P} = \frac{120(60)}{4} = 1800 \text{ rpm}$$

I

$$2. \quad n_m = (1-s)n_s \\ = (1-0.05) \times 1800 = 1710 \text{ rpm}$$

$$3. \quad f_r = s f_e = 0.05 \times 60 = 3 \text{ Hz}$$

$$4. \quad \tau_{load} = \frac{P_{out}}{\omega_m} = \frac{P_{out}}{2\pi \frac{n_m}{60}} \\ = \frac{10 \text{ hp} \times 746 \text{ watt / hp}}{1710 \times 2\pi \times (1/60)} = 41.7 \text{ Nm}$$



Induction Motors

A 208-V, 10hp, four pole, 60 Hz, Y-connected induction motor has a full-load slip of 5 percent

1. What is the synchronous speed of this motor?
2. What is the rotor speed of this motor at rated load?
3. What is the rotor frequency of this motor at rated load?
4. What is the shaft torque of this motor at rated load?

